Online PES-Adapted Neuromorphic Membrane–Gate Residual Control for Actuator-Constrained UAV Quaternion Attitude Tracking

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***Abstract*—We study a compact neuromorphic residual architecture for quaternion attitude tracking with bounded actuator commands. A fixed quaternion proportional– derivative (PD) controller is augmented by six paired continuous coupled membrane–gate (CMG) units whose gate-derived activities are decoded through eighteen weights adapted online by prescribed error sensitivity (PES). We distinguish intrinsic conductance modulation, decoder learning, and residual authority, and establish bounded neural states, bounded command deviation, projected decoder boundedness, and a restricted frozen-feature learning result. These properties do not imply closed-loop stability or saturation-free rotor allocation. In ten matched 45-s MuJoCo cases spanning five missions, a Skydio X2 model and a Crazyflie 2.0 (CF2) model with an explicit four-rotor actuation extension exhibit 17.3–60.6% lower mean quaternion-angle error than PD. Actuator-bound occupancy is zero except for brief Hardstep intervals. Direct and Nengo realizations agree within 2.013×10^{-6} in relative mean-error difference. Ablations identify online decoder adaptation, rather than conductance modulation, as the principal source of improvement in the tested amplified-gate regime. An annealed activity-trace learning variant is examined separately in the precursor architecture, with both beneficial and adverse outcomes. Preliminary physical 3DOF Hover Roll tests provide supporting evidence: three matched pairs improve their recorded mean error by $2.65\% \pm 1.08\%$ (sample standard deviation).**

I. Introduction

Learning-based augmentation can retain useful classical feedback structure while adapting a correction that is difficult to specify a priori. Residual robot control is established [1]; quadrotor examples include geometric $\mathcal{L}_1$ augmentation with performance guarantees [2] and PID-plus-PPO attitude residuals [3]. Thus, neither additive residual learning nor online flight adaptation is novel in isolation.

Neuromorphic control offers a complementary implementation approach. Nengo supports the Neural Engineering Framework (NEF) and online decoder learning [4]. PES-based adaptive control has been demonstrated on manipulators [5], [6], and a Nengo spiking PID controller with PES-based self-tuning has been validated on a free-flying UAV [7]. Our contribution is not the first PES/Nengo UAV controller. Instead, we investigate a specific continuous membrane–gate basis for a bounded quaternion-torque residual, with a fixed classical controller and adaptation confined to its decoder.

The central question is whether this small stateful basis can improve tracking, and which of its components explains the improvement. The paper contributes: (i) a code-aligned CMG–PES architecture with paired currents, explicit state/readout equations, and actuator interfaces (Fig. 1); (ii) precise boundedness and baseline-recovery properties, without an unsupported closed-loop stability claim; (iii) a five-mission, two-model study with direct/Nengo agreement and neuron-family comparisons; and (iv) a separate study of annealed activity-trace learning that documents its implementation and evidence limitations. Selected preliminary Hover tests support physical feasibility but are not the main validation claim.

Here "neuromorphic" refers to neuron-inspired continuous state dynamics, local error-directed learning, and a neural-simulator realization. The implementation does not emit spikes and has not been deployed on neuromorphic hardware; no power advantage is claimed. PES is an established rule, and novelty is claimed for its integration and controlled evaluation in this architecture, not for PES itself.

II. Simulation Models, Attitude Baseline, and Actuator Interface

A. MuJoCo Platforms and CF2 Actuation Extension

We use Crazyflie 2.0 (CF2) and Skydio X2 (X2) models derived from MuJoCo Menagerie [8]. Each vehicle has six rigid-body degrees of freedom: three translations and three rotations, represented by seven free-joint position coordinates (position and unit quaternion) and six velocity coordinates. The attitude controller requests three body torques; a separate classical height loop requests collective thrust. Thus attitude tracking is evaluated in a freely translating model, not an attitude-only joint model.

The upstream CF2 interface provides collective thrust and three abstract body-moment channels; its documentation identifies the original actuator limits as arbitrary and requiring tuning [9]. Our four-rotor actuation extension replaces that interface with four nonnegative, spatially located thrust inputs while retaining the source mass, inertia, and visual geometry.

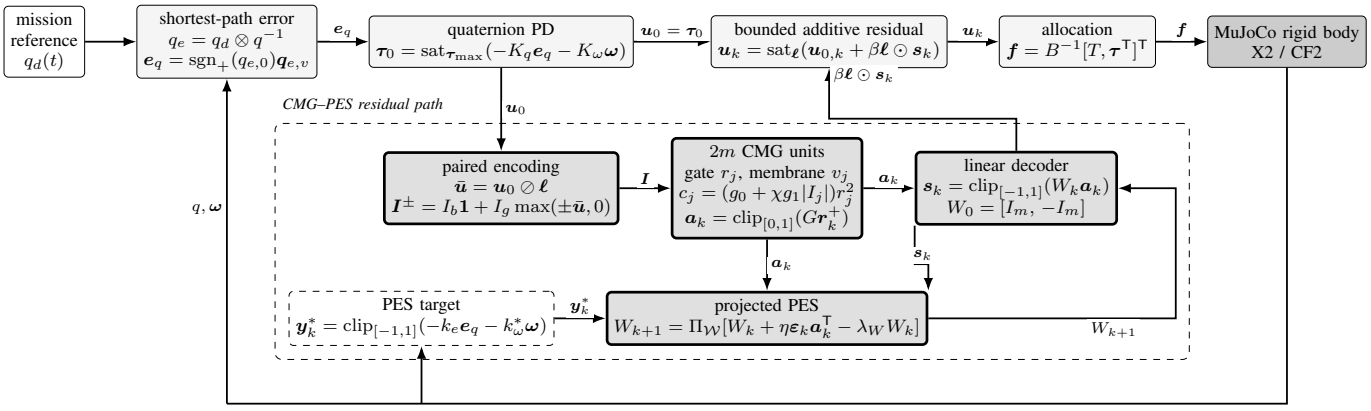


Fig. 1. CMG–PES residual architecture. The light blocks are the fixed classical path: error (6), PD law (12), and allocation (14) are never adapted. The heavy blocks encode the clipped baseline as paired currents (16), evolve $2m$ membrane–gate units (19), and decode a bounded residual (23) that is *added* to the saturated baseline (27). Only W_k adapts, through (26); $\beta = 0$ recovers the baseline exactly.



Fig. 2. Vehicles and simulation models. Left: CF2 hardware and MuJoCo model. Centre: X2 hardware and MuJoCo model. Right: both geometries at relative scale. The axis triads are body-frame axes: they are rigidly attached to the vehicle and therefore rotate with it relative to the inertial (world) frame (cf. Q: body $\to$ inertial in Sec. II).

Rotor moment arms and signed reaction-torque coefficients define an explicit allocation matrix, making the coupling between collective thrust, attitude torque, and individual actuator limits testable. This is a model-interface contribution, not a complete aerodynamic identification of CF2.

For CF2, the unscaled model's total thrust limit is 0.35 N. We evaluate a rotor-interface model with increased actuator authority: multiplying $f_{\max,i}$ by 1.5 yields 0.525 N total (Table II). This raises the modeled maximum thrust/weight from 1.321 to 1.982 at $g = 9.81$ m/s^2. For physical perspective, constant-height flight at a single-axis $45°$ tilt requires $T/(Mg) = 1/\cos 45° = \sqrt{2}$ even before reserving differential-thrust margin. The unscaled model cannot meet that idealized requirement; this does not establish such a limitation for stock CF2 hardware. The scale changes a simulation thrust bound, not rotor speed, stock propulsion capability, or tracking performance by 50%. Mass, inertia, and reaction coefficients are unchanged. PD and CMG use the same model and actuator bounds; any benefit from increased model authority is therefore distinct from the benefit of neural augmentation.

The masses agree with published specifications, but this does not validate all dynamic parameters (Table III). Bitcraze's historical measurement of 57.9 gram-force corresponds to approximately 0.568 N; the evaluated 0.525 N limit is about 7.5% below that operating point [12]. This supports a plausible thrust scale, not hardware validation, and does not establish that a stronger-than-stock motor is required. CF2 inertia is inherited from Menagerie, which cites system-identification work [9]; the rotor-extension moment arms and $|\kappa_i| = 0.005$ m remain design assumptions. X2 inertia, 13 N rotor limits, and $|\kappa_i| = 0.0201$ m are simulation parameters, not confirmed manufacturer characteristics.

Geometric dimensions also require like-for-like definitions. Bitcraze lists $92 \times 92 \times 29$ mm motor-to-motor/height dimensions [10]; our sites give an opposite-motor separation of 91.92 mm. Skydio lists an unfolded flying envelope of approximately $663 \times 569 \times 211$ mm [11], whereas the model's rotor centers span 280×360 mm. Center spacing and overall flying envelope are different quantities and do not by themselves establish geometric agreement or mismatch. Propulsion identification, motor and battery dynamics, and experimental validation remain necessary for hardware transfer; no endurance or electrical-power fidelity is claimed.

MuJoCo [14] integrates rigid-body motion, gravity, site-applied actuator wrenches, and enabled passive/contact forces. Both models specify air density 1.225 kg/m^3 and viscosity 1.8×10^{-5} Pa s; their default inertia-based fluid model [15] is a coarse drag approximation, not rotor-resolved aerodynamics. The runner overrides the timestep but retains each model's integrator. Actuators apply algebraic thrust commands without motor activation dynamics. Although gyro, accelerometer, and orientation sensors are defined, the evaluated feedback is read directly from `qpos`/`qvel`; optional measurement perturba-

TABLE I

NOMENCLATURE. PARENTHESISED TAGS GIVE THE DEFINING EQUATION; VECTORS ARE BOLD; $\mathrm{clip}_{[a,b]}$ AND $\mathrm{sat}_{\boldsymbol{\ell}}$ ACT ELEMENTWISE.

Attitude, plant, and notation					
$q,\ q_d$	state / reference quaternion	q_e	error quaternion, $q_d \otimes q^{-1}$	$q_{e,0}, \boldsymbol{q}_{e,v}$	scalar and vector parts
$\boldsymbol{e}_q$	shortest-path error (6)	θ_q	error angle (6)	$\boldsymbol{\omega}$	body angular rate
J	inertia tensor	Q	body-to-inertial rotation	$\otimes$	quaternion product
$\boldsymbol{\omega}_d, \boldsymbol{\omega}_e$	reference and error rate (8)	$R(\cdot)$	rotation matrix of a quaternion	$z, z_d, \dot{z}$	height, reference, rate (15)
$\odot, \oslash$	elementwise $\times$, $\div$	sgn_+	sign, with $\mathrm{sgn}_+(0) = 1$	σ	logistic function
$\mathbf{1}$	vector of ones	I_m	identity of order m	$\mathbb{R}$	real numbers
Fixed baseline and actuator interface					
$\boldsymbol{\tau}_0 = \boldsymbol{u}_0$	baseline torque (12)	K_q, K_ω	PD gains (12)	$\boldsymbol{\tau}_{\max}$	per-axis torque bound
ω_n, ζ	design frequency, damping	$\boldsymbol{\ell}$	per-channel command bound	$\mathrm{sat}_{\boldsymbol{\ell}}$	saturation at $\pm\boldsymbol{\ell}$
B	allocation matrix (13)	$\boldsymbol{f}$	rotor forces (14)	T	collective thrust (15)
κ_i	rotor reaction coefficient	m	torque channels, $m = 3$	M	vehicle mass
$\boldsymbol{f}_{\max}$	per-rotor force bound	g	gravity, $9.81\,\mathrm{m/s^2}$	n	runs per reported cell
Coupled membrane–gate unit					
v_j	membrane state (19)	r_j	gate state (19)	c_j	shunt conductance (19)
$\bar{\boldsymbol{u}}$	normalized baseline (16)	$\boldsymbol{I}^{\pm}, \boldsymbol{I}$	paired currents (16)	I_b, I_g	current offset and gain
τ, τ_r	membrane, gate time constants	v_{th}	gate threshold	$v_{\mathrm{rest}}, v_{\mathrm{reset}}$	rest level, reversal level
γ	gate self-coupling	α_h	logistic sharpness	χ	conductance switch, $\chi \in \{0, 1\}$
g_0, g_1	conductance gains (distinct from g)	G	activity gain, $G = 10^5$	h	integration step
$b_{j,k}$	gate target (21)	$r^+_{j,k}, v^+_{j,k}$	updated states (22)	$\lambda_{j,k}, d_{j,k}$	frozen membrane coefficients
Decoder, online learning, and residual authority					
$\boldsymbol{a}_k$	clipped activity (23)	W_k	decoder (23)	W_0	initial decoder, $[I_m, -I_m]$
$\boldsymbol{s}_k$	clipped readout (23)	$\boldsymbol{y}^*_k$	PES target (26)	$\boldsymbol{\varepsilon}_k$	PES error (26)
k_e, k^*_ω	target gains (26)	η	per-step learning rate	λ_W	per-step decoder decay
$\Pi_{\mathcal{W}}, W_{\max}$	projection and weight bound	β	residual authority (27)	$\boldsymbol{u}_k, \boldsymbol{\delta}_k$	applied command, deviation (27)
Annealed variant and reported metrics					
$\boldsymbol{z}_k$	activity trace (32)	$\rho(s)$	trace factor (32)	$s = t/T$	mission progress
η_0	initial annealed rate	$\bar{\theta}_q$	mean quaternion-angle error	S	actuator-bound occupancy [%]

TABLE II

COMPILED SIMULATION-MODEL CHARACTERISTICS. CF2 AUTHORITY IS SCALED RELATIVE TO THE UPSTREAM MODEL LIMIT, NOT A MANUFACTURER RATING. HARDWARE COMPARISONS ARE GIVEN IN TABLE III.

Property	CF2 rotor model	X2 model
Mass M (kg)	0.027	1.325
Rotor $\lvert p_x\rvert, \lvert p_y\rvert$ (m)	$0.0325, 0.0325$	$0.14, 0.18$
Reaction magnitude $\lvert\kappa_i\rvert$ (m)	0.005	0.0201
Unscaled rotor limit (N)	0.0875	13
Main-test authority scale	1.5	1
Main-test rotor limit (N)	0.13125	13
Main-test total limit (N)	0.525	52
Maximum thrust/weight	$1.321 \to 1.982$	4.001
Rigid-body integrator	RK4	Euler
Experiment timestep (s)	0.0005	0.0005
Motor activation/RPM states	None	None

TABLE III

SIMULATION PARAMETERS VERSUS MANUFACTURER INFORMATION. "NOT REPORTED" REFERS TO THE MANUFACTURER SOURCES REVIEWED, NOT PROOF THAT THE PARAMETER IS UNAVAILABLE ELSEWHERE.

Quantity	Simulation	Manufacturer information
CF2 mass	27 g	27 g [10]
X2 mass	1325 g	1325 g with battery [11]
CF2 total thrust	$0.350 \to 0.525$ N	0.568 N measured at 93.75% PWM[a] [12]
X2 rotor limit	13 N per rotor	Not reported [11]
CF2 motor response	No motor dynamics	~ 180 ms: $0 \to 25{,}000$ RPM[b] [13]
Inertia tensors	CF2 inherited; X2 geometry-derived	Not reported in product specifications [10], [11]

[a]Historical total-thrust measurement with a power-supply test setup, not a guaranteed continuous battery-powered rating. [b]One motor's reported full-command response with a charged battery; not an identified first-order time constant.

tions are separate runner settings. These experiments do not establish performance with a hardware state estimator.

B. Quaternion Convention and Error

Let $Q \in \mathbb{S}^3$ be the scalar-first Hamilton quaternion mapping body coordinates to inertial coordinates, and let $\boldsymbol{\omega}$ be body angular velocity. With physical inertia $\boldsymbol{J} \succ 0$ about the center of mass, expressed in body axes, the free-flight rotational balance is

$$\dot{Q} = \tfrac{1}{2} Q \otimes [0, \boldsymbol{\omega}^{\mathsf{T}}]^{\mathsf{T}}, \tag{1}$$

$$\boldsymbol{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_{\mathrm{act}} + \boldsymbol{\tau}_{\mathrm{p}} + \boldsymbol{\tau}_{\mathrm{d}} - \boldsymbol{\omega} \times (\boldsymbol{J}\boldsymbol{\omega}). \tag{2}$$

Here $\boldsymbol{\tau}_{\mathrm{p}}$ denotes modeled passive torque and $\boldsymbol{\tau}_{\mathrm{d}}$ any applied disturbance; the nominal ideal rigid-body model sets both to zero. Contact, if present, contributes an additional external wrench. The requested torque and actuator torque are distinct signals in Fig. 2.

The controller conjugates MuJoCo's orientation and uses the passive quaternion $q = Q^{-1}$, so

$$\dot{q} = -\tfrac{1}{2}[0, \boldsymbol{\omega}^{\mathsf{T}}]^{\mathsf{T}} \otimes q. \tag{3}$$

The desired q_d uses the same convention. Define

$$q_e = q_d \otimes q^{-1} = [q_{e,0}, \boldsymbol{q}_{e,v}^{\mathsf{T}}]^{\mathsf{T}}, \tag{4}$$

$$\boldsymbol{e}_q = \mathrm{sgn}_+(q_{e,0})\boldsymbol{q}_{e,v}, \qquad \mathrm{sgn}_+(0) = 1, \tag{5}$$

$$\theta_q = 2\,\mathrm{atan2}(\|\boldsymbol{q}_{e,v}\|_2, |q_{e,0}|) \in [0, \pi]. \tag{6}$$

The reported error θ_q is the shortest rotation angle, invariant under $q_e \mapsto -q_e$. Sign selection introduces a discontinuity at

$q_{e,0} = 0$; it does not establish robust hybrid global stability [16]. For a desired-frame rate $\boldsymbol{\omega}_d$ satisfying the corresponding passive kinematics, differentiation gives

$$\dot{q}_e = \tfrac{1}{2} q_e \otimes [0, \boldsymbol{\omega}_e^{\mathsf{T}}]^{\mathsf{T}}, \tag{7}$$

$$\boldsymbol{\omega}_e = \boldsymbol{\omega} - R(q_e)^{\mathsf{T}} \boldsymbol{\omega}_d, \qquad \dot{q}_{e,0} = -\tfrac{1}{2} \boldsymbol{q}_{e,v}^{\mathsf{T}} \boldsymbol{\omega}_e. \tag{8}$$

This is a kinematic identity, not a convergence result. Reference Euler triples parameterize q_d in the passive convention, rather than active body-to-world Euler commands.

C. Physical Inertia and Implemented PD Gains

MuJoCo stores principal moments $\boldsymbol{j}_{\mathrm{p}}$ in its inertial frame [17]. If R_{BI} maps that frame into body axes, the physical inertia is

$$\boldsymbol{J} = R_{BI} \operatorname{diag}(\boldsymbol{j}_{\mathrm{p}}) R_{BI}^{\mathsf{T}}. \tag{9}$$

The evaluated implementation uses a tuning vector $\boldsymbol{j}_{\mathrm{tune}} = \boldsymbol{j}_{\mathrm{p}}$ directly, in the returned array order:

$$\boldsymbol{j}_{\mathrm{tune}}^{\mathrm{CF2}} = (2.3951, 2.3951, 3.2347)^{\mathsf{T}} 10^{-5},$$
$$\boldsymbol{j}_{\mathrm{tune}}^{\mathrm{X2}} = (0.0607113, 0.0364684, 0.0254117)^{\mathsf{T}},$$

in $\mathrm{kg\,m^2}$. CF2's inertial axes coincide with its body axes. For X2 they do not; its physical body-frame tensor is approximately

$$\boldsymbol{J}_{\mathrm{X2}} = \begin{bmatrix} 0.0366517 & 0 & -0.0021000 \\ 0 & 0.0254117 & 0 \\ -0.0021000 & 0 & 0.0605280 \end{bmatrix} \ \mathrm{kg\,m^2}. \tag{10}$$

Thus the tuning vector must not be identified with $\operatorname{diag}(\boldsymbol{J})$ for both vehicles.

The fixed quaternion PD baseline is

$$\boldsymbol{k}_q = 2\boldsymbol{j}_{\mathrm{tune}} \odot \boldsymbol{\omega}_n^2, \qquad \boldsymbol{k}_\omega = 2\zeta \boldsymbol{j}_{\mathrm{tune}} \odot \boldsymbol{\omega}_n, \tag{11}$$

$$\boldsymbol{u}_0 = \mathrm{sat}_{\boldsymbol{\ell}}(-\boldsymbol{k}_q \odot \boldsymbol{e}_q - \boldsymbol{k}_\omega \odot \boldsymbol{\omega}), \tag{12}$$

where $\odot$ denotes componentwise multiplication, $\boldsymbol{\omega}_n = (4, 4, 3.5)^{\mathsf{T}}$ rad/s, $\zeta = 0.95$, and $\mathrm{sat}_{\boldsymbol{\ell}}$ clips component i to $[-\ell_i, \ell_i]$. For these rotor-interface models, the model-policy torque limits are $\ell_i = \max(j_{\mathrm{tune},i} a_i, 10^{-6}\ \mathrm{N\,m})$, with $\boldsymbol{a} = (12, 12, 10)^{\mathsf{T}}$ rad/s^2. These are controller torque bounds, not guaranteed feasible rotor torques. The law is quaternion feedback [18]; $\boldsymbol{\omega}_n$ parameterizes its gains but is not a verified set of closed-loop natural frequencies for coupled X2 dynamics. Desired-rate feedforward and gyroscopic cancellation are omitted. The identical fixed gains are retained for PD and its CMG augmentation.

For perspective only, take a constant reference, scalar proportional gain $\boldsymbol{k}_q = k_q \mathbf{1}$, the ideal plant ($\boldsymbol{\tau}_{\mathrm{p}} = \boldsymbol{\tau}_{\mathrm{d}} = 0$), no saturation or residual, and a trajectory confined to one smooth sign branch. Then $V = \frac{1}{2}\boldsymbol{\omega}^{\mathsf{T}} \boldsymbol{J} \boldsymbol{\omega} + 2k_q(1 - |q_{e,0}|)$ gives $\dot{V} = -\boldsymbol{\omega}^{\mathsf{T}} \operatorname{diag}(\boldsymbol{k}_\omega) \boldsymbol{\omega}$. This restricted identity does not prove stability of the anisotropic, saturated, moving-reference experiments.

D. Requested and Realized Wrench

For rotor positions $\boldsymbol{p}_i$ in body coordinates, parallel $+z$ thrust axes, and signed reaction coefficients κ_i, define

$$B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ p_{1,y} & p_{2,y} & p_{3,y} & p_{4,y} \\ -p_{1,x} & -p_{2,x} & -p_{3,x} & -p_{4,x} \\ \kappa_1 & \kappa_2 & \kappa_3 & \kappa_4 \end{bmatrix}. \tag{13}$$

The signs follow $\boldsymbol{p}_i \times [0, 0, f_i]^{\mathsf{T}}$ and the actuator's local reaction torque $\kappa_i f_i$. Both models have zero body-frame center-of-mass x, y offsets, so the site's x, y coordinates also give the relevant center-of-mass moment arms. Figure 2 fixes the rotor ordering used by B; κ_i has units of meters, not angular velocity. The runner computes

$$\begin{aligned} \boldsymbol{f}_{\mathrm{req}} &= B^{-1}[T, \boldsymbol{u}^{\mathsf{T}}]^{\mathsf{T}}, \\ \boldsymbol{f}_{\mathrm{act}} &= \mathrm{clip}_{[\mathbf{0}, \boldsymbol{f}_{\max}]}(\boldsymbol{f}_{\mathrm{req}}), \\ [T_{\mathrm{act}}, \boldsymbol{\tau}_{\mathrm{act}}^{\mathsf{T}}]^{\mathsf{T}} &= B \boldsymbol{f}_{\mathrm{act}}. \end{aligned} \tag{14}$$

Here $\boldsymbol{u} = \boldsymbol{u}_0$ for PD; the CMG path in Section III modifies that requested torque before the same allocation and clipping. Each f_i is a commanded force in newtons, not a rotor-speed command. Exact wrench realization requires $B^{-1}[T, \boldsymbol{u}^{\mathsf{T}}]^{\mathsf{T}} \in [\mathbf{0}, \boldsymbol{f}_{\max}]$. A componentwise torque box is not generally contained in this collective-dependent feasible set. Clipped inverse allocation is used, not an optimization-based allocator [19].

Collective thrust remains classical, with $z_d = 1$ m:

$$T = \mathrm{clip}_{[0, 2.5Mg]} \left(\frac{M[g + 6(z_d - z) - 4\dot{z}]}{\max(R(Q)_{33}, 0.35)} \right). \tag{15}$$

This expression uses the model's $g = 9.81$ m/s^2. The height loop reads free-joint height and vertical velocity; the denominator compensates tilt with a lower guard. Its upper bound does not supersede individual rotor limits. The CMG path has no direct collective output, but can affect height through attitude, tilt compensation, and allocation saturation. The resulting shared plant interface isolates changes in requested attitude torque while preserving the same physical simulation for both controllers.

III. CMG State Dynamics and Online Decoder Adaptation

A. Encoding, States, and Numerical Update

Symbols are collected in Table I. For m torque channels and $2m$ units, normalize the clipped baseline and form positive/negative current pairs:

$$\begin{aligned} \bar{\boldsymbol{u}} &= \boldsymbol{u}_0 \oslash \boldsymbol{\ell}, \\ \boldsymbol{I}^{\pm} &= I_b \mathbf{1} + I_g \max(\pm \bar{\boldsymbol{u}}, \mathbf{0}), \\ \boldsymbol{I} &= [(\boldsymbol{I}^{+})^{\mathsf{T}}, (\boldsymbol{I}^{-})^{\mathsf{T}}]^{\mathsf{T}}. \end{aligned} \tag{16}$$

Each unit has a dimensionless membrane state v_j and gate r_j:

$$\dot{r}_j = [\sigma(\alpha_h(v_j - v_{\mathrm{th}} + \gamma r_j)) - r_j]/\tau_r, \tag{17}$$

$$\dot{v}_j = -(v_j - v_{\mathrm{rest}})/\tau + I_j/\tau - c_j(v_j - v_{\mathrm{reset}}), \tag{18}$$

$$c_j = (g_0 + \chi g_1 |I_j|) r_j^2, \qquad \chi \in \{0, 1\}. \tag{19}$$

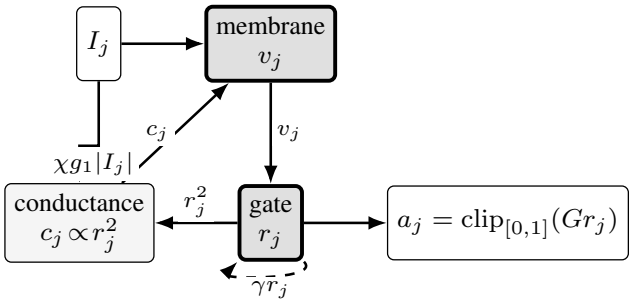


Fig. 3. Coupling topology of one CMG unit, cf. (19). The membrane sets the gate target; the gate acts back twice, shifting its own threshold through γ and shunting the membrane through $c_j \propto r_j^2$ ($\chi = 1$ makes that shunt input dependent). The dynamics are continuous: no spike, no reset, and only a_j leaves the unit.

The logistic gate target depends on v_j, while r_j shifts that target and modulates membrane conductance (Fig. 3). Thus, both states participate even though the decoder reads the gate-derived activity. The model has no threshold-triggered reset or spike event; v_{reset} is a reversal level in a continuous conductance term. The gains g_0, g_1 have the units needed to make c_j a rate.

The actual integration is sequential, not exponential Euler for both states. From the old states compute

$$b_{j,k} = \sigma\Big(\mathrm{clip}_{[-60,60]}[\alpha_h(v_{j,k} - v_{\mathrm{th}} + \gamma r_{j,k})]\Big), \tag{20}$$

$$r^+_{j,k} = \mathrm{clip}_{[0,1]}[r_{j,k} + (h/\tau_r)(b_{j,k} - r_{j,k})]. \tag{21}$$

Then evaluate $c_{j,k} = (g_0 + \chi g_1 |I_{j,k}|)(r^+_{j,k})^2$, $\lambda_{j,k} = \tau^{-1} + c_{j,k}$, and $d_{j,k} = (v_{\mathrm{rest}} + I_{j,k})/\tau + c_{j,k} v_{\mathrm{reset}}$. The frozen-coefficient membrane step is

$$v^+_{j,k} = e^{-\lambda_{j,k}h} v_{j,k} + (1 - e^{-\lambda_{j,k}h}) d_{j,k}/\lambda_{j,k}. \tag{22}$$

This exactly solves the scalar membrane equation with those frozen coefficients; it does not establish unconditional stability of the coupled nonlinear closed loop. Assign $(v_{j,k+1}, r_{j,k+1}) = (v^+_{j,k}, r^+_{j,k})$.

B. Readout, PES, and Additive Residual

The within-step activity, initial decoder, and clipped readout are

$$\begin{aligned} \boldsymbol{a}_k &= \mathrm{clip}_{[0,1]}(G\boldsymbol{r}^+_k), \qquad W_0 = [I_m, -I_m], \\ \boldsymbol{s}_k &= \mathrm{clip}_{[-1,1]}(W_k \boldsymbol{a}_k). \end{aligned} \tag{23}$$

Pair subtraction retains directional information from a nonnegative activity basis; state memory does not imply instantaneous sign agreement after a command reversal. The teaching target and local update are

$$\boldsymbol{y}^*_k = \mathrm{clip}_{[-1,1]}(-k_e \boldsymbol{e}_{q,k} - k^*_\omega \boldsymbol{\omega}_k), \tag{24}$$

$$\boldsymbol{\varepsilon}_k = \boldsymbol{y}^*_k - \boldsymbol{s}_k, \tag{25}$$

$$W_{k+1} = \Pi_{\mathcal{W}}[W_k + \eta \boldsymbol{\varepsilon}_k \boldsymbol{a}_k^{\mathsf{T}} - \lambda_W W_k]. \tag{26}$$

Here $\mathcal{W} = [-W_{\max}, W_{\max}]^{m \times 2m}$, and η and λ_W are per-step coefficients, not continuous-time rates. Holding them fixed while changing h changes the learning dynamics. PES learns a prescribed residual target; it does not differentiate through the plant or directly minimize the closed-loop tracking cost. In particular, with a clipped readout this rule is not the exact gradient of the clipped instantaneous squared loss. It is a PES-style error-directed update with projection and leak.

Using W_k before the update, the requested command is

$$\boldsymbol{u}_k = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{u}_{0,k} + \beta \boldsymbol{\ell} \odot \boldsymbol{s}_k), \quad \boldsymbol{\delta}_k = \boldsymbol{u}_k - \boldsymbol{u}_{0,k}, \qquad 0 \le \beta \le 1. \tag{27}$$

The canonical UAV case has $m = 3$, six CMG units, and eighteen decoder weights. Intrinsic conductance modulation (χ), synaptic/decoder adaptation (W_k), and output authority ($\beta\boldsymbol{\ell}$) are distinct.

C. Structural Properties and Their Scope

Proposition 1 (Bounded internal states)**.** *With the nonnegative conductance parameters in Table IV, $r_{j,0} \in [0,1]$, and $0 < h \le \tau_r$, the gate remains in $[0,1]$. The membrane iterate remains in the convex hull of its initial value, v_{reset}, and the interval $[v_{\mathrm{rest}} + I_b, v_{\mathrm{rest}} + I_b + I_g]$.*

Proof. The unclipped gate step is a convex combination of $r_{j,k}$ and $b_{j,k} \in [0,1]$. Since $I_j \in [I_b, I_b + I_g]$ and $c_j \ge 0$, d_j/λ_j is a convex combination of $v_{\mathrm{rest}} + I_j$ and v_{reset}. Equation (22) is a convex combination of $v_{j,k}$ and d_j/λ_j. Induction proves both statements. □

Proposition 2 (Bounded authority and baseline recovery)**.** *For $\boldsymbol{u}_{0,k} \in \mathcal{U}$ and any finite W_k, $\boldsymbol{u}_k \in \mathcal{U}$ and $|\delta_{k,i}| \le \beta \ell_i$. If $\beta = 0$, the requested baseline command is recovered at every step. If $W_k = 0$ at a particular step, it is recovered at that step, but learning need not keep $W_{k+1} = 0$.*

Proof. The outer clip enforces membership in $\mathcal{U}$. Its componentwise nonexpansiveness and $\mathrm{sat}_{\ell_i}(u_{0,k,i}) = u_{0,k,i}$ give $|\delta_{k,i}| \le |\beta \ell_i s_{k,i}| \le \beta \ell_i$. The recovery statements follow by substitution. □

The same allocator and state then give the same actuator output when $\beta = 0$. The proposition does not establish state safety, rotor headroom, stability, or a bound on tracking improvement. A bounded adversarial correction can still degrade or destabilize a nonlinear plant.

Proposition 3 (Decoder bound and restricted contraction)**.** *If $W_0 \in \mathcal{W}$, then* (26) *keeps $W_k \in \mathcal{W}$. If, additionally, the target and a nonzero activity $\boldsymbol{a}$ are fixed, leak is zero, and readout clipping and projection remain inactive, then*

$$\boldsymbol{\varepsilon}_{k+1} = (1 - \eta \|\boldsymbol{a}\|_2^2) \boldsymbol{\varepsilon}_k. \tag{28}$$

For a nonzero initial error, its norm contracts to zero when $0 < \eta \|\boldsymbol{a}\|_2^2 < 2$.

Proof. Projection gives the first result. Under the additional conditions, $(W_{k+1} - W_k)\boldsymbol{a} = \eta \boldsymbol{\varepsilon}_k \|\boldsymbol{a}\|_2^2$; subtracting this from the fixed target error gives (28). The scalar multiplier has modulus less than one under the stated condition. □

This is the constant-feature PES result [20], not a closed-loop convergence theorem. Since $\|\boldsymbol{a}\|^2 \le 2m$, $0 < \eta < 1/m$ is sufficient for that restricted model with nonzero activity; changing targets, loss of excitation, clipping, and leakage invalidate a direct application to the experiment. No unique identification of all decoder weights follows from one fixed regressor.

D. Annealed Activity-Trace Variant

The eprop anneal replaces instantaneous presynaptic activity with an exponentially smoothed trace. We call it *annealed activity-trace learning* (AAT), not full e-prop: it does not compute the recurrent state-sensitivity eligibility derivatives of [21]. For $\xi_k = \mathrm{clip}(t_k/T, 0, 1)$,

$$\rho_k = 0.98 - 0.48\xi_k, \qquad \boldsymbol{z}_{-1} = 0, \tag{29}$$

$$\boldsymbol{z}_k = \rho_k \boldsymbol{z}_{k-1} + (1-\rho_k)\boldsymbol{a}_k, \tag{30}$$

$$\eta_k^{\mathrm{AAT}} = 2\eta_{\mathrm{PES}}(1 - 0.8\xi_k), \tag{31}$$

$$W_{k+1} = \Pi_{\mathcal{W}}[W_k + \eta_k^{\mathrm{AAT}} \boldsymbol{\varepsilon}_k \boldsymbol{z}_k^{\mathsf{T}} - \lambda_W W_k]. \tag{32}$$

The factor two is present in the current experiments. Thus this comparison changes both temporal smoothing and learning-rate schedule. At $h = 0.5$ ms, the instantaneous frozen-ρ equivalent time constant $-h/\log\rho$ falls from about 24.75 ms to 0.72 ms; it does not become exactly instantaneous. Meanwhile the learning rate falls from $2\eta_{\mathrm{PES}}$ to $0.4\eta_{\mathrm{PES}}$.

Proposition 4 (Trace bound and frozen-regressor interpretation)**.** *With $\boldsymbol{z}_{-1} = 0$, $\boldsymbol{a}_k \in [0,1]^{2m}$, and $0 \le \rho_k \le 1$, the trace stays in $[0,1]^{2m}$. Projection keeps AAT decoder weights in $\mathcal{W}$. For fixed target and fixed $\boldsymbol{a}, \boldsymbol{z}$, with no leak, readout clipping, or projection, its error obeys $\boldsymbol{\varepsilon}_{k+1} = (1 - \eta_k^{\mathrm{AAT}} \boldsymbol{z}^{\mathsf{T}}\boldsymbol{a})\boldsymbol{\varepsilon}_k$.*

Proof. Trace boundedness follows from convexity. Decoder boundedness follows from projection. The last identity follows by multiplying the unprojected weight increment by $\boldsymbol{a}$ and subtracting it from the target error. □

Hence alignment between trace and present activity affects learning. This frozen-regressor identity does not establish convergence when the trace, reference, and plant evolve together.

E. Nengo Realization

In Nengo 4.1.0, CMG is a custom non-spiking `NeuronType` with probeable membrane and gate states. Constructed currents are injected into `ensemble.neurons`; conventional NEF gain/intercept fitting is bypassed. The result is a direct-current neural basis with a Nengo PES decoder, not a claim that the full classical controller has been compiled into an NEF population representation.

With unfiltered activity and $n = 2m$, Nengo's update scaling is matched using $\eta_N = \eta n/h$ and Nengo error $\boldsymbol{s}_k - \boldsymbol{y}_k^*$. The implementation subtracts $\lambda_W W_k$ using the pre-step weights and projects after the simulator step. The measured worst relative mean-error difference across the ten missions is 2.013×10^{-6}; an isolated command test gives approximately 1.05×10^{-7}. These are numerical comparisons over tested trajectories, not bitwise identity or independent validation of the control hypothesis.

TABLE IV
CANONICAL 3-CHANNEL CMG–PES PARAMETERS. NEURAL STATES/CURRENTS ARE NORMALIZED MODEL VARIABLES; LEARNING COEFFICIENTS ARE PER STEP.

Quantity	Value
h, duration	0.0005 s, 45 s
τ, τ_r	0.05 s, 0.005 s
$v_{\mathrm{rest}}, v_{\mathrm{reset}}, v_{\mathrm{th}}$	$0, 0, 1$
$v_{j,0}, r_{j,0}, \chi$	$0, 0, 1$
$\alpha_h, \gamma, g_0, g_1$	$50, 2, 120, 30$
I_b, I_g, G	$0.75, 1.25, 10^5$
$\beta, \eta, \lambda_W, W_{\max}$	$0.5, 8\times10^{-4}, 10^{-6}, 6$
k_e, k_ω^*	$8, 0$

IV. Experimental Protocol and Cross-Mission Results

A. Models, Parameters, and Measurements

MuJoCo supplies rigid-body and actuator simulation [14]. The X2 model uses four rotor actuators. The CF2 case is an explicit four-rotor design with $1.5\times$ nominal upper thrust authority; it is neither stock-Crazyflie hardware validation nor a validated motor upgrade. For both rotor models, $\boldsymbol{\ell} = \max(\mathrm{diag}(J)\odot(12,12,10)^{\mathsf{T}}, 10^{-6}\mathbf{1})$ in N m, with the maxima taken componentwise.

Each baseline/CMG pair uses the same initial pose/rate, reference, duration, allocator, model, and timestep. The primary campaign is nominal and deterministic, with no injected disturbances or sensor noise. One run per cell is reported; repeated numerical integration of the same deterministic case is not statistical replication. Learning remains active throughout evaluation. We report the sample mean $\bar\theta_q$ and relative reduction $100(\bar\theta_{q,\mathrm{PD}} - \bar\theta_{q,\mathrm{CMG}})/\bar\theta_{q,\mathrm{PD}}$. Summary metrics use all 90,000 integration samples; most saved plotting traces retain every tenth sample (5-ms spacing), while the X2 Hard45 trace is full-rate. Thus recomputing a mean from the decimated plot data need not exactly reproduce the full-rate metric.

The five reference triples are generated as follows (radians): Roll is $(0.25\sin 0.7t, 0, 0)$; Full3 is $0.5\sin(t + (0,\pi/2,\pi)^{\mathsf{T}})$, componentwise. Hard45 is

$$\begin{bmatrix} 0.55\sin(0.75t) + 0.20\sin(2.10t + 0.40) \\ 0.45\sin(0.95t + \pi/3) + 0.18\sin(1.75t) \\ 0.65\sin(0.55t + \pi) + 0.25\sin(1.35t + 0.20) \end{bmatrix}. \tag{33}$$

Hardstep holds zero for one second, then cycles every two seconds through the rows of

$$\begin{bmatrix} .75 & -.45 & .55 \\ -.65 & .55 & -.70 \\ .50 & .50 & .90 \\ -.80 & -.35 & .20 \end{bmatrix}. \tag{34}$$

Yaw360 is $(0, 0, 2\pi(3\xi^2 - 2\xi^3))$ with $\xi = t/45$. The nominal Hover equilibrium is also available, but its essentially zero error does not support a meaningful percentage comparison.

TABLE V
CANONICAL NENGO CMG–PES VERSUS PD: MEAN QUATERNION-ANGLE ERROR [RAD], 45 S, $n = 1$ PER CELL. S IS THE LOGGED PERCENTAGE OF SAMPLES WITH AT LEAST ONE ROTOR COMMAND AT A BOUND, PD/CMG. CF2 DENOTES THE 1.5× ROTOR-AUTHORITY DESIGN, NOT STOCK HARDWARE.

Model	Mission	PD	CMG–PES	Reduction	S [%]
X2	Roll	.050565	.041785	17.36%	0/0
	Full3	.303275	.131748	56.56%	0/0
	Hard45	.294082	.138007	53.07%	0/0
	Hardstep	.454528	.359355	20.94%	.016/.239
	Yaw360	.075863	.029902	60.58%	0/0
CF2	Roll	.051337	.042439	17.33%	0/0
	Full3	.288383	.126476	56.14%	0/0
	Hard45	.285449	.132319	53.65%	0/0
	Hardstep	.426182	.347886	18.37%	.137/.264
	Yaw360	.075812	.029870	60.60%	0/0

B. Primary Five-Mission Matrix

All ten primary cases improve mean error (Table V). Gains are largest on Yaw360 and Full3 and smallest on Roll; the abrupt Hardstep case is intermediate. The reported bound-occupancy statistic detects post-clipping commands within the runner's tolerance of a limit. It is not a measurement of requested-wrench infeasibility magnitude. Hardstep has small but nonzero occupancy on both models, so a blanket zero-saturation claim would be incorrect. These cases demonstrate repeatable nominal performance, not robustness to model mismatch or a universally optimal PD comparison. Figures 4 and 5 show the X2 Hard45 case, the largest smooth-mission gain, together with the neural activity that produced it; Fig. 6 shows the Hardstep case, where the reduction is smallest and the only nonzero bound occupancy occurs.

C. What Is Learning?

Fixed-decoder residuals already improve some missions: in the audited direct X2 runs their mean reductions range from about 0.12% to 10.27%; the corresponding CF2-design range is 0.06% to 6.80%. Thus it is inaccurate to say that fixing the decoder removes *all* improvement. PES produces the much larger 17.3–60.6% primary reductions. Separately, toggling χ with the decoder mode held fixed is numerically indistinguishable under the tested settings. This identifies decoder adaptation as the principal tested lever, not a universal claim that conductance never matters.

Because $a_j = \mathrm{clip}(10^5 r_j, 0, 1)$, decoder activity saturates already at $r_j = 10^{-5}$. Tiny but nonzero gates can therefore yield order-one activity while $c_j \propto r_j^2$ remains small. This is a plausible mechanistic explanation for the observed conductance insensitivity, not a claim that every gate remains $O(10^{-6})$ at every instant. A calibrated gate-gain sweep and gate/membrane readout ablation are needed to determine whether CMG offers benefits beyond a simpler dynamical basis.

Figure 7 shows the CF2 Yaw360 case, the largest reduction in Table V, and Fig. 8 distinguishes the membrane, raw gate, and clipped activity using that same recorded trajectory rather than equating them.

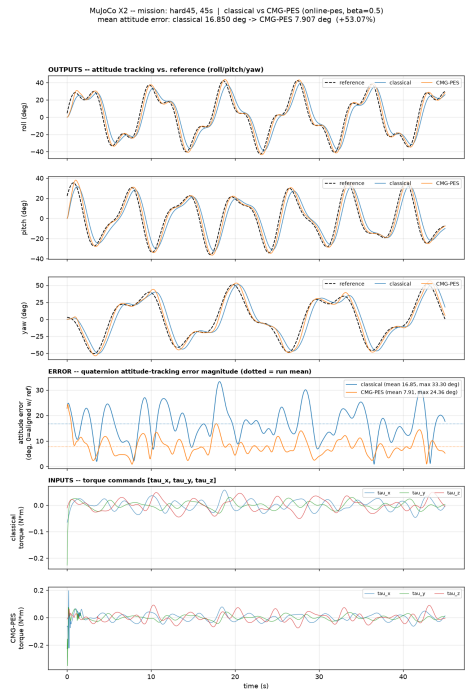


Fig. 4. X2 Hard45: classical baseline vs. CMG–PES residual control.

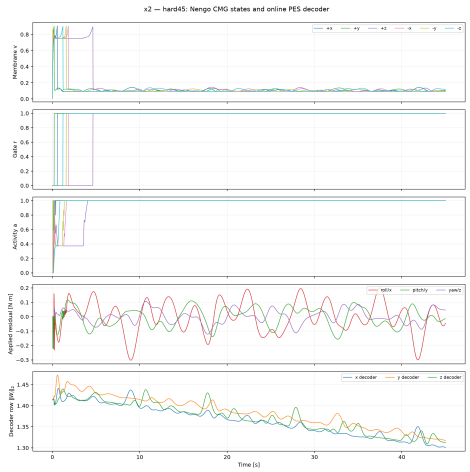


Fig. 5. X2 Hard45: neural activity for the CMG residual controller.

D. Neuron-Family Comparison

The Nengo benchmark contains two models, five missions, and eleven controllers: PD and fixed/PES variants of CMG, LIF, ALIF, Izhikevich [22], and LIFRate. All use the same residual/allocator path, with model-specific native activity and filtering. Some Izhikevich–PES cells outperform CMG–PES on mean tracking error. The audit reports an approximate average normalized residual RMS of 0.37 for Izhikevich–PES versus 0.12 for CMG–PES, alongside more decoder movement. Equal output caps do not imply equal realized correction effort. Very sparse LIF/ALIF activity in some tested cells is an operating-point limitation, not evidence that those neuron

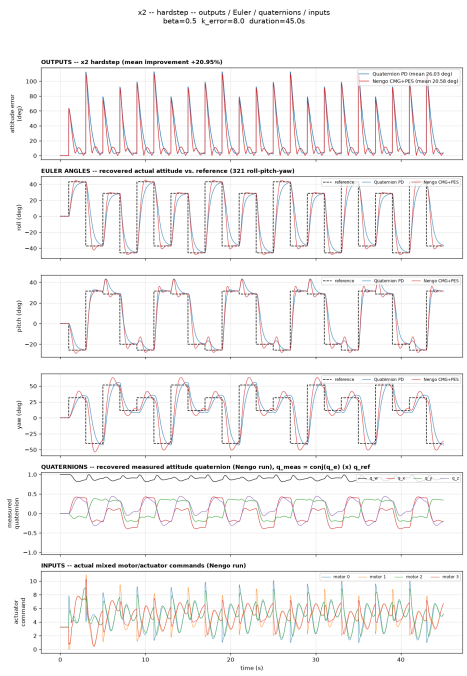


Fig. 6. X2 Hardstep: attitude tracking time series (hard step reference).

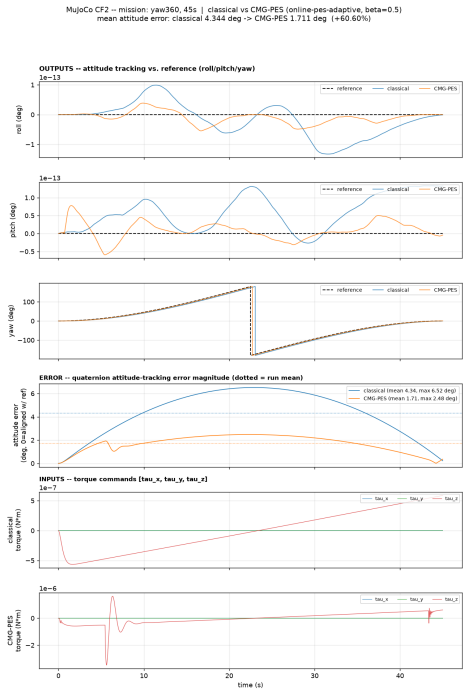


Fig. 7. CF2 Yaw360: classical baseline vs. CMG–PES residual control.

families cannot learn. Accordingly, the supported conclusion is a tracking–effort tradeoff, not universal CMG superiority or a proven causal effect of spiking.

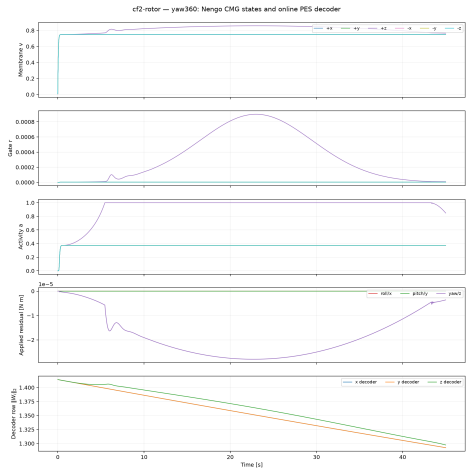


Fig. 8. CF2 Yaw360: neural activity for the CMG residual controller.

TABLE VI
TRACEABLE PRECURSOR SUMMARIES: MEAN QUATERNION-ANGLE ERROR [RAD], 45 S, $h = 0.5$ MS, ONE RUN PER CONFIGURATION. AAT CORRESPONDS TO ARCHIVED `EPROP_ANNEAL`. THESE RESULTS MUST NOT BE POOLED WITH TABLE V.

Model	Mission	PD	PES	AAT
X2	Hard45	.294079	.122051	.127526
CF2	Hard45	.285448	.035549	.035445
CF2	Hardstep	.661866	1.757814	2.079473

V. ANNEALED LEARNING: VERIFIED PRECURSOR EVIDENCE

The verified AAT results use the precursor *blend* law $(1-\beta)\boldsymbol{u}_0+\beta\boldsymbol{\ell}\odot\boldsymbol{s}$ at $\beta = 0.5$, with ideal external-wrench injection and a classical horizontal stabilizer. Its CF2 torque limits also differ from the rotor-design campaign. Therefore these results are a within-precursor learning-configuration comparison, not additional rotor-feasible validation of (27).

On Hard45, AAT is slightly better than PES for CF2 and worse for X2 (Table VI). Both learned configurations severely degrade CF2 Hardstep tracking relative to that precursor baseline. Since quaternion angle is bounded by π, these values indicate a loss of tracking, not mathematical divergence of the angle or decoder weights. The archived Hardstep saturation field is zero and the actuation path is ideal wrench; these files do not establish rotor-allocation saturation as the cause.

The comparison changes eligibility smoothing and learning-rate schedule together, and the canonical experiment additionally changes residual law and actuator interface. It therefore does not isolate which change avoids the precursor failure. At fixed $\beta = 0.5$, the blend retains a known baseline coefficient of 0.5, not an arbitrarily vanishing coefficient. The additive law removes this explicit attenuation but cannot by itself guarantee stability or eliminate adverse step responses. An additive-versus- blend factorial study with matched actuation, teacher, and learning rate is the appropriate next test.

VI. Selected Preliminary Physical Tests

Before the present manuscript synthesis, CMG–PES was tested on the Quanser 3DOF Hover. These are preliminary supporting tests, not free-flight validation of either MuJoCo airframe. The rig uses torque-to-motor-voltage allocation, a 200-Hz Q8 loop, and ten 0.5-ms Nengo substeps per hardware sample. The representative Roll configuration uses $\beta = 0.2$, a 0.2-V per-motor residual cap, a 1 V/s residual slew limit, $\eta = 2 \times 10^{-4}$, $k_e = 2$, and $k_\omega^* = 0$.

Roll is the strongest replicated result: over three matched 8-s pairs, the recorded mean error is $5.4405 \pm 0.0648^\circ$ (PD) versus $5.2960 \pm 0.0428^\circ$ (CMG–PES), i.e., a paired mean reduction of $2.65\% \pm 1.08\%$ (sample SD, not a confidence interval). These archived values use the mean norm of the wrapped yaw/pitch/roll error vector (not $\bar{\theta}_q$) and are not pooled with the MuJoCo table.

Selected later single pairs show mean-error reductions of 0.55% (Hard45) and 0.95% (scaled, transition-smoothed Hard-step) while worsening peak errors. Other profiles regress, and no full-scale yaw360 pair was validated. Thus hardware results support execution feasibility and a small repeated Roll benefit, but do not explain the simulation-to-hardware discrepancy.

VII. Discussion and Conclusion

We combine paired continuous CMG dynamics, local online decoder adaptation, and a bounded additive torque correction around a fixed quaternion-PD law. Across five missions (ten matched 45-s cases), mean attitude error decreases by 17.3–60.6% on two modeled platforms with explicit actuator-bound accounting; Nengo reproduces the direct implementation, and ablations isolate decoder learning as the dominant effect in the tested regime. AAT provides a well-defined alternative memory/schedule mechanism, but its verified results remain precursor evidence rather than a cross-mission advantage.

Bounded commands, internal states, and projected weights are implementation properties, not stability/safety proofs. Establishing ISS/Lyapunov results would require baseline robustness assumptions and bounds on the *realized* post-allocation perturbation; small voltage caps alone do not quantify achievable improvement. Follow-up work includes matched-rate AAT/PES comparisons, equal-effort neuron/readout ablations, stochastic disturbance trials, frozen-weight held-out tests, repeated hardware perturbations, and neuromorphic-device energy/latency measurements.

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